

13

MEASUREMENT AND SPACE

SIMILAR FIGURES

Unique to Australia are a number of tourist attractions known as the 'Bigs' or 'Biggies'.

- the Big Merino (Goulburn, NSW)
- the Big Banana (Coffs Harbour, NSW)
- the Big Golden Guitar (Tamworth, NSW)
- the Big Pineapple (Woombye, Qld)
- the Big Orange (Berri, SA)
- the Big Crocodile (Wyndham, WA)
- the Big Captain Cook (Cairns, Qld)
- the Big Mango (Bowen, Qld)
- the Big Rocking Horse (Gumeracha, SA)
- the Giant Koala (Dadsells Bridge, Vic)

Each attraction is a scale model or enlargement of an Australian object.
The Big Pineapple is 16 metres high.



Chapter outline

		Working mathematically				
13.01	Similar figures	U	F		R	C
13.02	Finding sides in similar figures	U	F	PS	R	
13.03	Scale drawings	U	F	PS	R	

Wordbank

enlargement An increase in the size of a shape

image A transformed shape after it has been enlarged or reduced

matching angles corresponding angles of similar figures in the same position

matching sides corresponding sides of similar figures in the same position

scale factor The amount by which a shape has been enlarged or reduced, equal to $\frac{\text{image length}}{\text{original length}}$

similar To have the same shape but not necessarily the same size, an enlargement or reduction (symbol: ~)

superimpose To place one figure on top of another figure so that sides and angles match

Quiz
Wordbank 13

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (2):

13.02 Finding an unknown side in similar figures

13.03 Scale drawings

***Twig* videos (5):**

13.03 Jai Singh (sundial) • Modelling the Spitfire • Queen Hatshepsut's ship • The history of the golden ratio • Fractals: The Koch snowflake

Quizzes (5):

- Wordbank 13
- SkillCheck 13
- Mental skills 13
- Language of maths 13
- Test yourself 13

Skillsheet (1):

13.02 Finding sides in similar triangles

Worksheets (8):

13.01 A page of similar figures • A page of congruent and similar figures • Enlargements and reductions • Enlarging a logo

13.02 Finding sides in similar figures

13.03 Problems involving scale drawings • Investigating paper sizes

Mind map Similar figures

Puzzles (3):

13.01 Cartoon enlargement

13.02 Similar triangles • Nested similar triangles

Technology (1):

13.03 Converting map scales to ratios

Spreadsheet (1):

13.03 Converting map scales to ratios

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

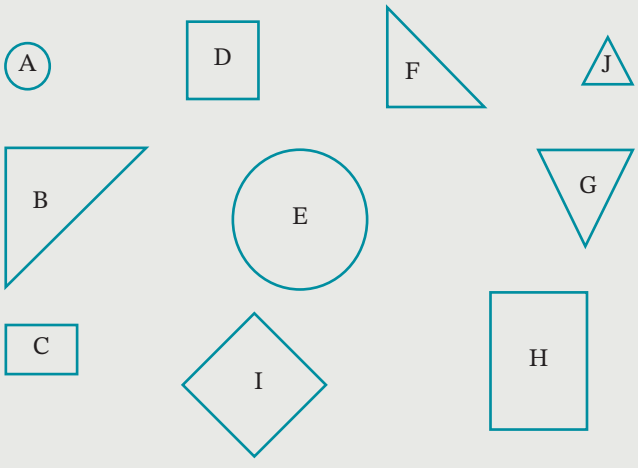
In this chapter you will:

- identify and use similar figures
- investigate the properties of similar figures
- use scale factors to find unknown sides in similar figures
- measure and calculate using scale drawings

Quiz
SkillCheck 13

SkillCheck ANSWERS ON P. 606

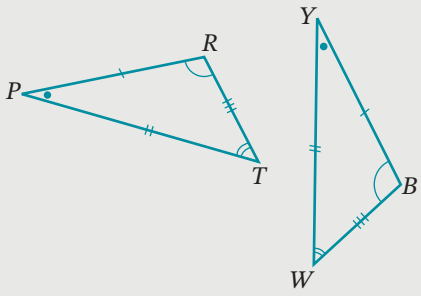
1 Identify pairs of shapes where one shape is an enlargement of the other.



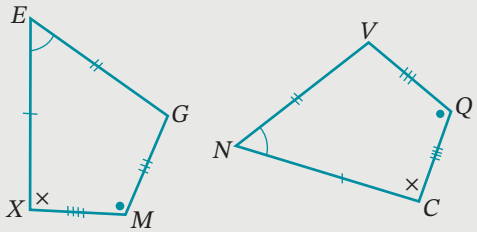
2 For each pair of identical figures, list all pairs of:

i matching angles **ii** matching sides

a



b



3 Identify pairs of equivalent fractions from this list.

$\frac{52}{13}$ $\frac{15}{20}$ $\frac{8}{12}$ $\frac{64}{16}$ $\frac{24}{36}$ $\frac{11}{44}$ $\frac{18}{24}$ $\frac{8}{32}$

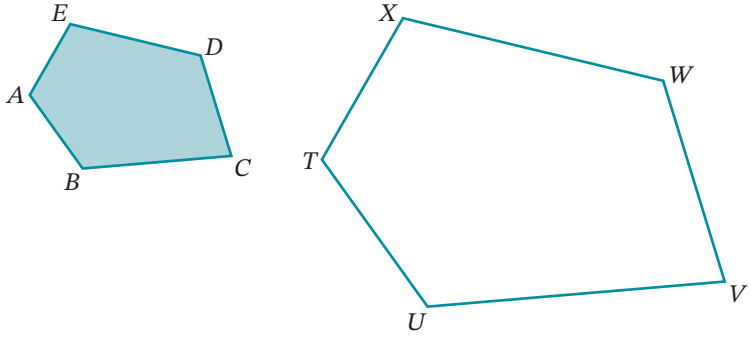
4 Solve each equation.

a $\frac{x}{24} = \frac{5}{8}$ **b** $\frac{m}{7} = \frac{10}{14}$ **c** $\frac{a}{2} = \frac{4}{8}$ **d** $\frac{k}{15} = \frac{2}{3}$

e $\frac{7}{12} = \frac{x}{3}$ **f** $\frac{5}{6} = \frac{n}{4}$ **g** $\frac{t}{2} = \frac{7}{3}$ **h** $\frac{d}{5} = \frac{17}{10}$

Investigation

Same shape, different size



1 **a** Consider the pentagons *ABCDE* and *TUVWX*. Do they look similar?

b What do you think is meant by the word 'similar'?

c Copy and complete this table by measuring each angle and side in the 2 'similar' pentagons.

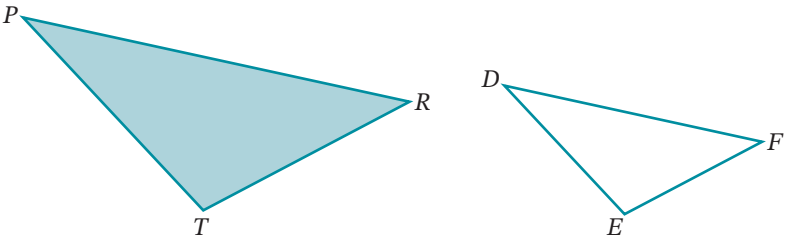
d What do you notice about the sizes of the matching angles?

e What do you notice about the sizes of the matching sides?

Angles		Sides (in mm)	
$A =$	$T =$	$AB =$	$TU =$
$B =$	$U =$	$BC =$	$UV =$
$C =$	$V =$	$CD =$	$VW =$
$D =$	$W =$	$DE =$	$WX =$
$E =$	$X =$	$AE =$	$TX =$

Foundation Standard Complex

2 $\triangle PTR$ has been reduced to produce the similar triangle $\triangle DEF$.



a List the 3 pairs of matching angles and measure the size of each one.

b List the 3 pairs of matching sides and measure the length of each one.

3 Copy and complete:

a For similar figures, m_____ angles are e_____.

b Similar figures have the s_____ shape, but not the same s_____.

13.01 Similar figures

Worksheets
A page of similar figures
A page of congruent and similar figures
Enlargements and reductions
Enlarging a logo

Puzzle
Cartoon enlargement

Similar figures have the same shape but are not necessarily the same size.

When a photograph is enlarged or reduced, all parts of the picture remain the same shape, but all lengths and distances will be respectively larger or smaller than the corresponding lengths and distances on the original.

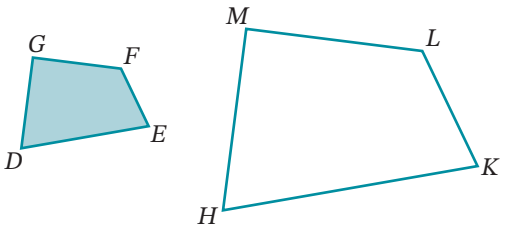


iStock.com/aluxum

When a figure is enlarged or reduced, a **similar figure** is created. The original figure is called the **original**, while the enlarged or reduced figure is called the **image**.

The scale factor

The **scale factor** describes by how much a figure has been enlarged or reduced. For example, if the quadrilateral *DEFG* is enlarged by doubling each side, the scale factor is 2. Quadrilaterals *DEFG* and *HKLM* are similar. $\angle D$ matches $\angle H$, $\angle E$ matches $\angle K$, and so on.



Original Image

$HK = 2 \times DE$

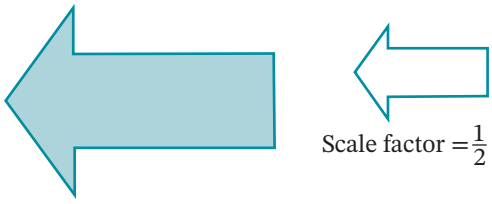
$KL = 2 \times EF$

$LM = 2 \times FG$

$HM = 2 \times DG$

$\therefore$ Scale factor $= 2$

In a similar manner, if the arrow figure is reduced by halving each side, the scale factor is $\frac{1}{2}$.



Original Image

Scale factor

$$\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$$

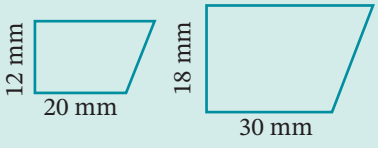
- If the scale factor is greater than 1, then the image is an enlargement.
- If the scale factor is between 0 and 1, then the image is a reduction.

Example 1

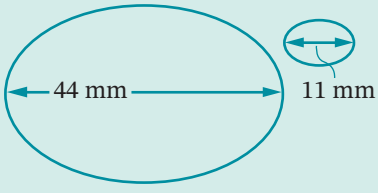
Find the scale factor for each pair of similar figures.

In all questions, assume the left figure is the original and the right figure the image.

a



b



Solution

a Scale factor $= \frac{30}{20}$ (or $\frac{18}{12}$) $\quad \frac{\text{image length}}{\text{original length}}$

$= 1\frac{1}{2}$

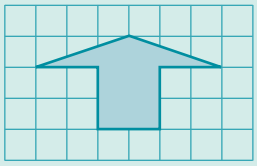
b Scale factor $= \frac{11}{44}$ $\quad \frac{\text{image length}}{\text{original length}}$

$= \frac{1}{4}$

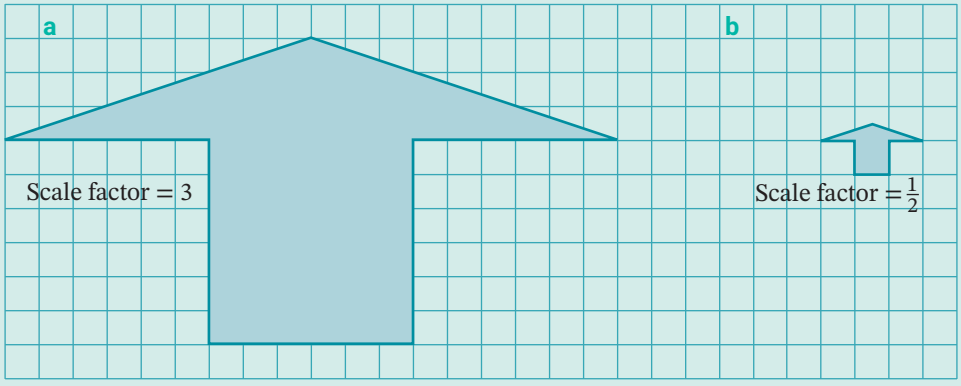
Example 2

Construct the image of the figure shown if the scale factor is:

a 3 **b** $\frac{1}{2}$



Solution



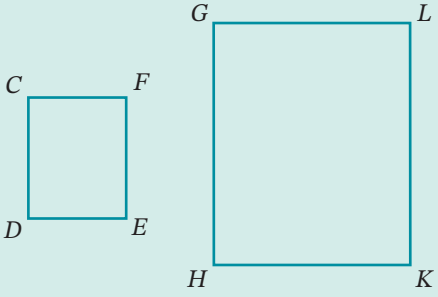
The similarity symbol ~

The symbol for 'is similar to' is '~'. We must make sure that the vertices (or angles) of the similar figures are written in matching order.

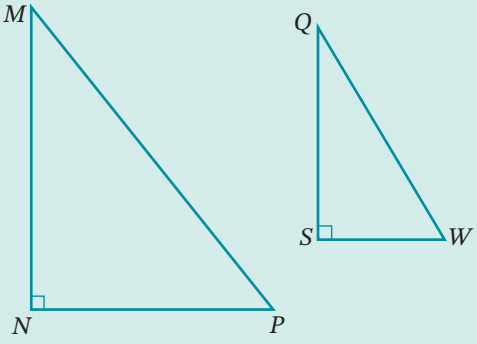
Example 3

Test whether each pair of figures are similar by measuring them (in millimetres).

a



b



Solution

a The rectangle $GHKL$ is an enlargement of rectangle $CDEF$.

$$\text{Scale factor} = \frac{GH}{CD} = \frac{32}{16} = 2$$

$$\frac{HK}{DE} = \frac{26}{13} = 2$$

$$\therefore \frac{GH}{CD} = \frac{HK}{DE}$$

$\therefore CDEF \sim GHKL$ Matching order of vertices

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b $\triangle QWS$ is not a reduction of $\triangle MPN$.

$$\frac{QS}{MN} = \frac{28}{40} = \frac{7}{10}$$

$$\frac{SW}{NP} = \frac{16}{32} = \frac{1}{2}$$

$$\frac{QS}{MN} \neq \frac{SW}{NP}$$

$\therefore \triangle MNP$ is not similar to $\triangle QSW$.

Matching sides of similar figures are **corresponding sides** that are in the same position.

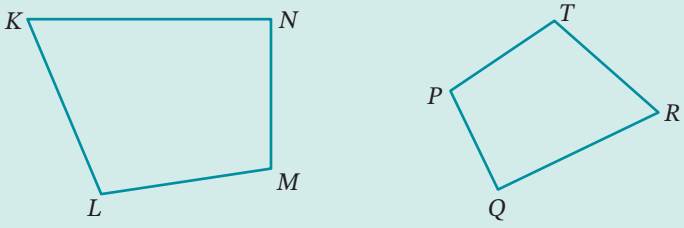
Matching angles of similar figures are **corresponding angles** that are in the same position.

One way of testing whether 2 figures are similar is to **superimpose** one shape onto the other, that is, to move it to a position on top of the other so that sides and angles match.

Similar figures may be identified by superimposing them through a combination of translations (slides), rotations (turns) and reflections (flips).

Example 4

The 2 quadrilaterals *KLMN* and *PQRT* are similar.



a List all pairs of matching sides and matching angles.

b Use the correct notation to write a similarity statement relating these 2 quadrilaterals.

Solution

a By rotating the figure *KLMN* and superimposing it over *PQRT*, its shape can be matched with *PQRT*.

The pairs of matching sides are:	The pairs of matching angles are:
KN and *QR*	$\angle K$ and $\angle R$
MN and *PQ*	$\angle N$ and $\angle Q$
ML and *PT*	$\angle M$ and $\angle P$
LK and *TR*.	$\angle L$ and $\angle T$.

b $\angle K$ matches with $\angle R$, $\angle L$ matches with $\angle T$, $\angle M$ matches with $\angle P$, $\angle N$ matches with $\angle Q$.

$\therefore KLMN \sim RTPQ$ Matching order of vertices

EXERCISE 13.01 ANSWERS ON P. 606

Similar figures UFRC

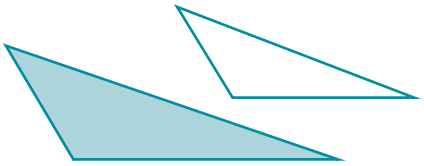
EXAMPLE 1

1 By measurement, find the scale factor for each pair of similar figures.

a



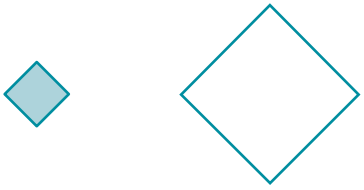
b



c



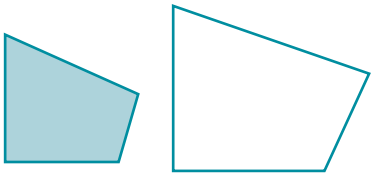
d



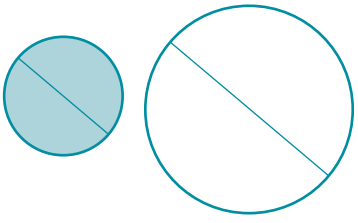
e



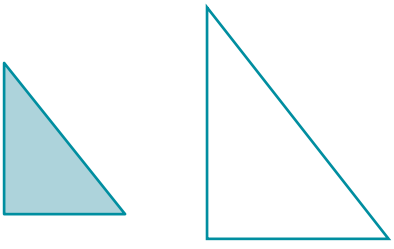
f



g



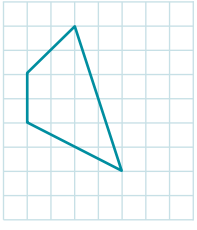
h



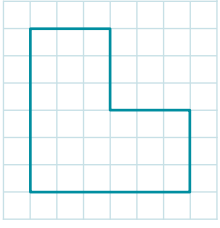
EXAMPLE 2

2 Copy each figure onto graph paper and draw its image using the given scale factor.

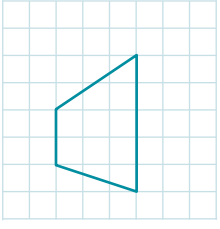
a Scale factor = 2



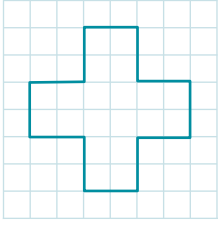
b Scale factor = $\frac{1}{3}$



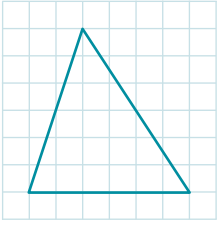
c Scale factor = 2



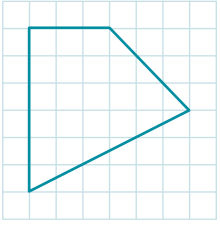
d Scale factor = 3



e Scale factor = $\frac{1}{2}$



f Scale factor = $\frac{2}{3}$



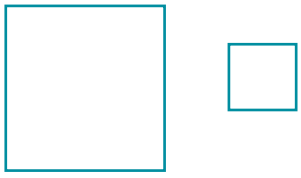
Foundation Standard Complex

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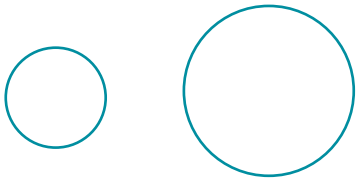
3 Are each pair of figures similar?

EXAMPLE 3

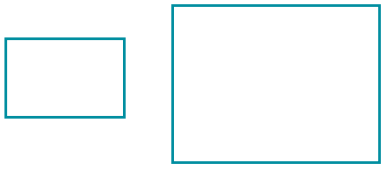
a



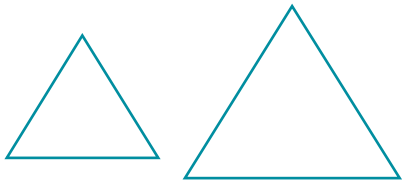
b



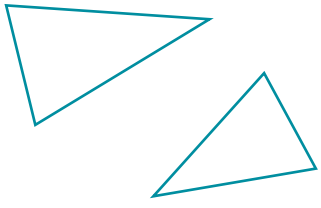
c



d



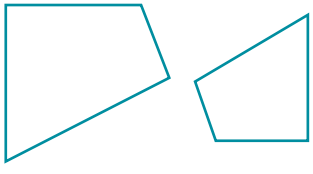
e



f



g



h



4 For each pair of similar figures: R C

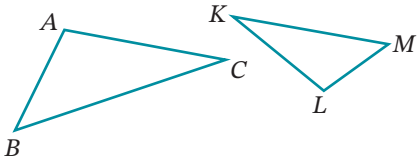
EXAMPLE 4

i list all pairs of matching angles

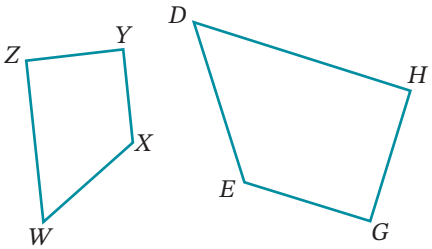
ii list all pairs of matching sides

iii use the correct notation ('~') to write a similarity statement relating them.

a

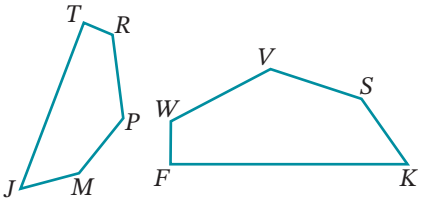


b

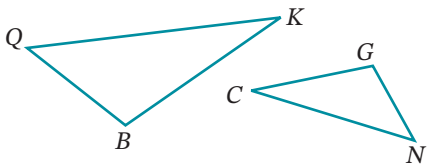


Foundation | Standard | Complex

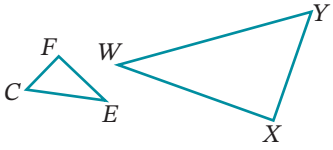
c



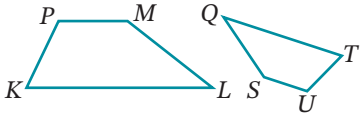
d



e



f



Quiz
Mental skills 13

Mental skills 13: Maths without calculators ANSWERS ON P. 607

Simplifying fractions and ratios

When simplifying a fraction or a ratio, look for a common factor to divide into both the numerator and the denominator, preferably the highest common factor (HCF). Remember the divisibility tests:

A number is divisible by ...	if ...
2	it is even (its last digit is 2, 4, 6, 8 or 0)
3	the sum of its digits is divisible by 3
4	its last 2 digits form a number divisible by 4
5	its last digit is 0 or 5
6	it is even and the sum of its digits is divisible by 3
9	the sum of its digits is divisible by 9
10	its last digit is 0

1 Study each example.

a Simplify $\frac{27}{45}$.

$\frac{\cancel{27}^{9}}{\cancel{45}_{15}} = \frac{9}{15}$ dividing numerator and denominator by 3

$\frac{\cancel{9}^{3}}{\cancel{15}_{5}} = \frac{3}{5}$ dividing numerator and denominator by 3 again

Note: This fraction could be simplified in one step if you divided by 9, the highest common factor (HCF) of 27 and 45.

b Simplify $\frac{160}{400}$.

$\frac{160}{400} = \frac{16}{40}$ dividing numerator and denominator by 10

$\frac{16^{2}}{40_{5}} = \frac{2}{5}$ dividing numerator and denominator by 8

Note: This fraction could be simplified in one step if you divided by 80, the HCF of 160 and 400.

c Simplify 24 : 36.

$24 : 36 = 24^{4} : 36^{6} = 4 : 6$ dividing both terms by 6

$4^{2} : 6^{3} = 2 : 3$ dividing both terms by 2

Note: This fraction could be simplified in one step if you divided by 12, the HCF of 160 and 400.

d Simplify 135 : 90.

$135 : 90 = 135^{27} : 90^{18} = 27 : 18$ dividing both terms by 5

$27^{3} : 18^{2} = 3 : 2$ dividing both terms by 9

e Calculate $\frac{3}{8} \times \frac{2}{15}$ in simplest form.

$\frac{3}{8} \times \frac{2}{15} = \frac{3}{{}_{4}8} \times \frac{2^{1}}{15}$ dividing 2 and 8 by 2

$= \frac{{}^{1}3}{{}_{4}8} \times \frac{2^{1}}{15_{5}}$ dividing 3 and 15 by 3

$= \frac{1}{20}$

f What fraction is 36 minutes of 1 hour?

$\frac{36}{1\text{ h}} = \frac{36\text{ min}}{60\text{ min}} = \frac{3}{5}$

2 Now simplify each fraction or ratio.

a $\frac{10}{15}$	**b** $\frac{16}{20}$	**c** $\frac{30}{42}$	**d** $\frac{8}{16}$
e $\frac{20}{80}$	**f** $\frac{6}{36}$	**g** $\frac{20}{24}$	**h** $\frac{12}{30}$
i 20 : 36	**j** 25 : 45	**k** 18 : 40	**l** 28 : 35
m 27 : 21	**n** 16 : 12	**o** $\frac{5}{6} \times \frac{18}{25}$	**p** $\frac{12}{50} \times \frac{10}{21}$

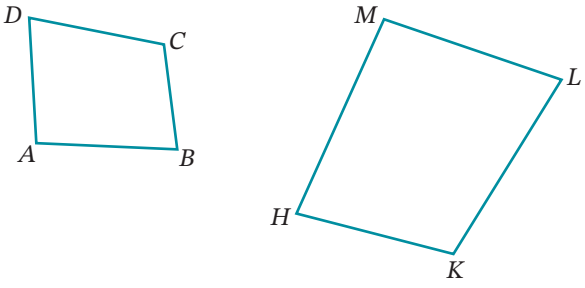
3 Express each as a simplified fraction.

a 425 g of 1 kg	**b** 8 months of 1 year	**c** 64 cm of 1 m
d 750 mL of 3 L	**e** 10 hours of 2 days	**f** 80c of $10

Investigation

Properties of similar figures

1 The quadrilaterals *ABCD* and *HKLM* are similar.



a Measure the sizes of each pair of matching angles. What do you notice?

b Measure the lengths of each pair of matching sides.

c Calculate each ratio of matching sides, as a decimal correct to one decimal place.

i $\frac{MH}{AB}$ **ii** $\frac{HK}{BC}$ **iii** $\frac{KL}{CD}$ **iv** $\frac{ML}{AD}$

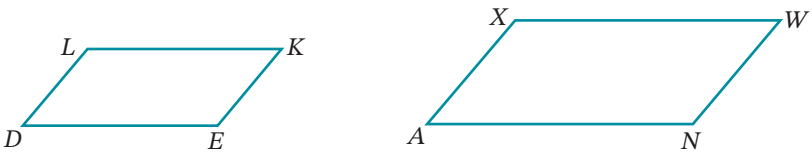
What do you notice?

d Copy and complete.

i The matching a_____ of similar figures are e_____.

ii Matching s_____ of similar figures are in the same r_____.

2 Consider the parallelograms *DEKL* and *ANWX*.



a Measure the sizes of each pair of matching angles. Are they equal?

b Measure the lengths of each pair of matching sides.

c Calculate each ratio of matching sides, as a decimal correct to one decimal place:

i $\frac{AN}{DE}$ **ii** $\frac{AX}{DL}$

Are they equal?

d *DEKL* is not similar to *ANWX*. Explain why.

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Finding sides in similar figures

13.02

Properties of similar figures

- **Matching angles** are equal
- **Matching sides** are in the same ratio

Skillsheet
Finding sides in similar triangles

13.02

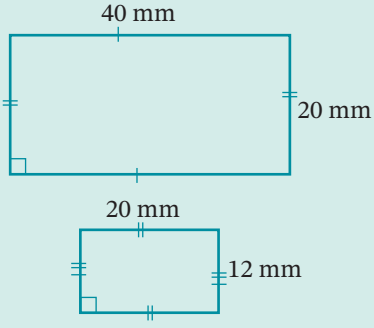
Worksheet
Finding sides in similar figures

Puzzle
Similar triangles

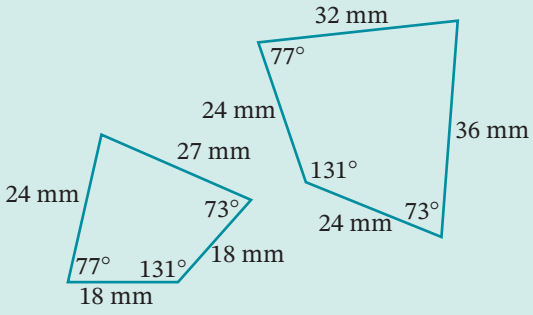
Example 5

Test whether each pair of figures are similar.

a



b



Solution

a For the 2 rectangles, matching angles are equal (90°), but the ratios of the matching sides are not equal $\left(\frac{20}{40}=\frac{1}{2} \text{ but } \frac{12}{20}=\frac{3}{5}\right)$.
∴ The rectangles are not similar.

b For the 2 quadrilaterals, matching angles are equal and the ratios of matching sides are equal $\left(\frac{32}{24}=\frac{36}{27}=\frac{24}{18}=\frac{4}{3}\right)$.
∴ The quadrilaterals are similar.

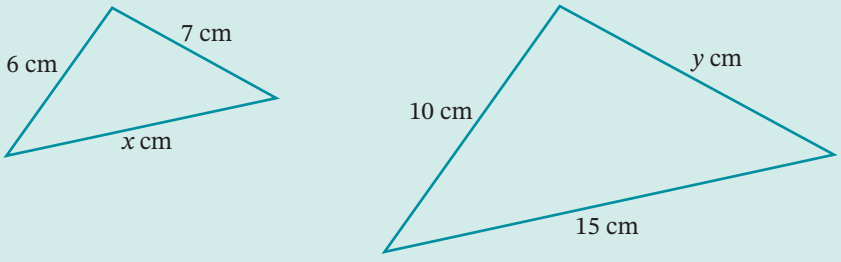


iStock.com/ivanastar

Video
Finding an unknown side in similar figures

Example 6

These 2 triangles are similar. Find the value of x and y.



Solution

Since the triangles are similar, the ratios of matching sides are equal.

$\frac{x}{15} = \frac{6}{10}$

$x = \frac{6}{10} \times 15$

$= 9$

$\frac{y}{7} = \frac{10}{6}$

$y = \frac{10}{6} \times 7$

$= 11\frac{2}{3}$

Alternative method:

Scale factor $= \frac{10}{6} = \frac{5}{3}$

$x = 15 \div \frac{5}{3}$

$= 9$

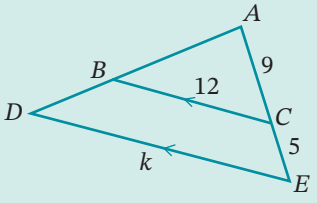
$y = 7 \times \frac{5}{3}$

$= 11\frac{2}{3}$

Puzzle
Nested similar triangles

Example 7

$\triangle ABC \sim \triangle ADE$. Find the value of k.



Solution

$\frac{DE}{BC} = \frac{AE}{AC}$ Ratios of matching sides are equal

$\frac{k}{12} = \frac{14}{9}$ $AE = 9 + 5 = 14$

$k = \frac{14}{9} \times 12$

$= 18\frac{2}{3}$

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EXERCISE 13.02 ANSWERS ON P. 607

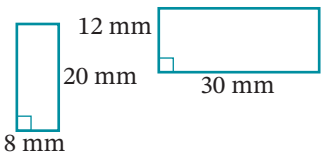
Finding sides in similar figures U F PS R

EXAMPLE 5

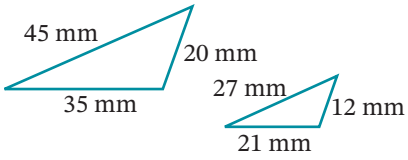
13.02

1 Test whether each pair of figures are similar. R

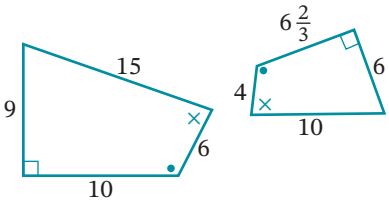
a



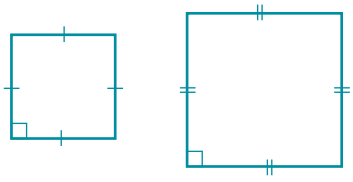
b



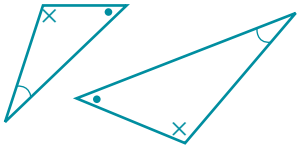
c



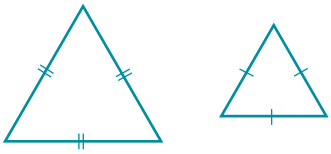
d



e



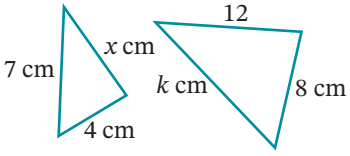
f



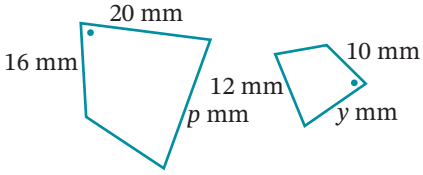
EXAMPLE 6

2 Find the value of each variable in each pair of similar figures. R

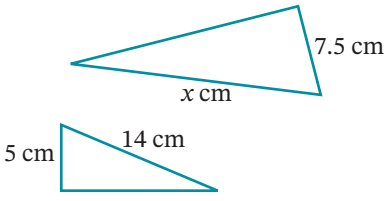
a



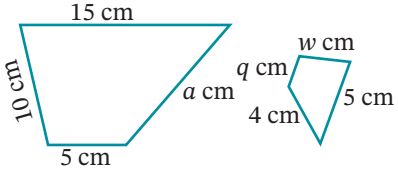
b



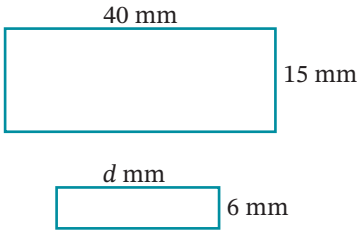
c



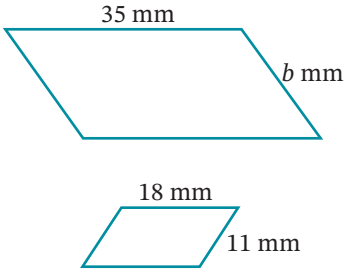
d



e

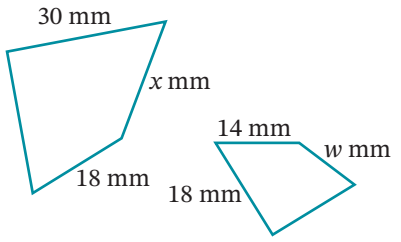


f

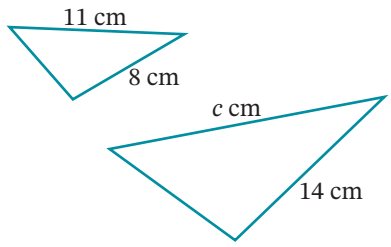


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g

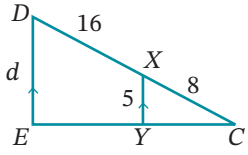


h

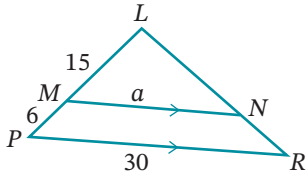


EXAMPLE 7

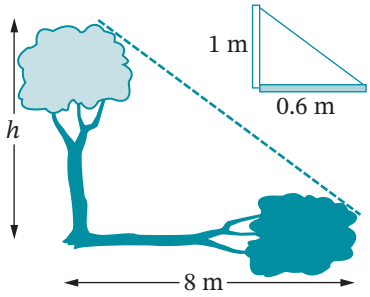
3 $\triangle CED \sim \triangle CYX$. Find the value of d. R



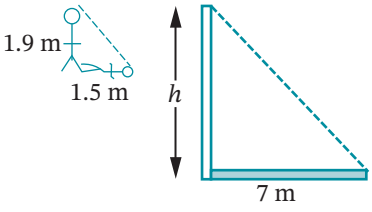
4 $\triangle LMN \sim \triangle LPR$. Find the value of a. R



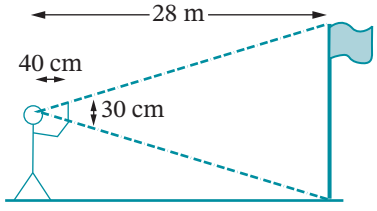
5 A metre stick casts a shadow of length 0.6 m when a tree casts a shadow of 8 m. Find the height h of the tree. R



6 Ayon, who is 1.9 m tall, casts a 1.5 m long shadow. At the same time, a light pole casts a shadow 7 m long. What is the height of the light pole, correct to one decimal place? R



7 Emily holds up a small 40 cm long stick from her eye so that it appears to be the same height as the flagpole. She is standing 28 m from the flagpole and the stick is 30 cm long. Find the height of the flagpole. PS R

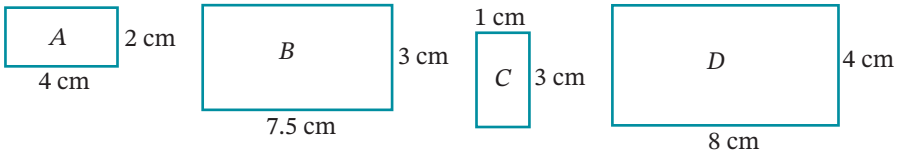


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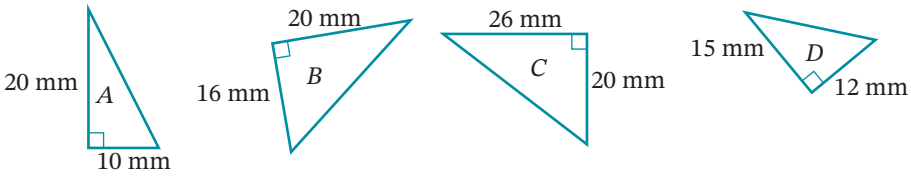
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8 For each set of figures, find the pair of similar figures and the scale factor between them. R

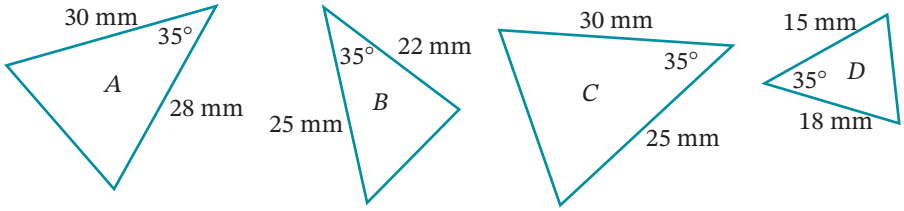
a



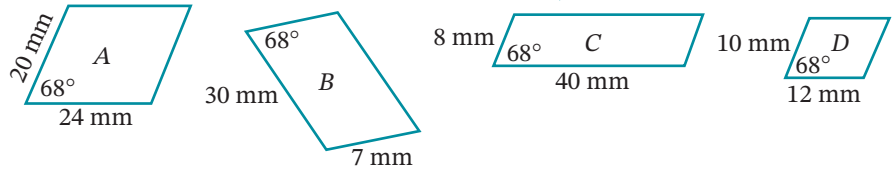
b



c



d



Investigation

Are all shapes similar?

Work in groups of 3 or 4 to complete this activity.

You will need: a ruler, a pencil, scissors.

1 **a** Draw 2 equilateral triangles, using a different side length for each triangle.

b Are the 2 equilateral triangles similar? Give reasons.

c Compare your triangles with those of other members of your group. Are all equilateral triangles similar? Give reasons.

2 **a** Draw a pair of similar isosceles triangles.

b Are all isosceles triangles similar? Give reasons.

3 **a** Draw a pair of rectangles that are similar.

b Draw a pair of rectangles that are *not* similar.

c When are rectangles similar? Discuss your answer with other students.

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4 State whether the following statements are true or false. Give reasons for your answers. (Making a few sketches will help.)

a All parallelograms are always similar.

b All circles are always similar.

c All squares are always similar.

d All rhombuses are always similar.

13.03 Scale drawings

Worksheet
Problems involving scale drawings

Technology
Converting map scales to ratios

Spreadsheet
Converting map scales to ratios

A **scale drawing** accurately represents a larger or smaller object such that the lengths and distances on it are in the same ratio as the real lengths and distances. This means that the scale drawing is either an enlargement or reduction of the real object. The scale factor used is called the **scale** of the diagram.

A scale may be represented in different ways.

- 1 cm = 10 m or 1 cm to 10 m (a pair of corresponding measurements)
- 1 : 50 (a ratio)
- 

Scaled length

The scale on a scale drawing is written as a ratio of **scaled length : real length**, where **scaled length** is the length on the diagram.

Videos
Scale drawings

Jai Singh (sundial)

Modelling the Spitfire

Queen Hatshepsut's ship

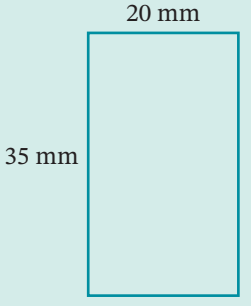
The history of the golden ratio

Fractals: The Koch snowflake

For example, a scale of 1 : 100 means that the real lengths are 100 times larger than the lengths on the diagram.

Example 8

A scale drawing of a rectangular block of land has been drawn. What scale has been used?





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Solution

By measurement, the length and width of the scale drawing are 35 mm and 20 mm.

Scale = 35 mm : 35 m (or 20 mm : 20 m) scaled length : real length

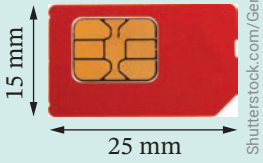
= 35 mm : 35 000 mm (or 20 mm : 20 000 m) 1 m = 1000 mm

= 1 : 1000

The actual length and width are 1000 times the scaled length and width.

Example 9

A SIM card for a mobile phone has a length of 25 mm and a width of 15 mm. A scale diagram of the SIM card is shown below.



Shutterstock.com/GeniusKp

By measurement, determine what scale has been used.

Solution

By measurement, the length and width of the scale diagram are 75 mm and 45 mm respectively.

Scale = 75 mm : 25 mm (or 45 mm : 15 mm) scaled length : real length

= 3 : 1

The actual length and width are $\frac{1}{3}$ of the scaled length and width.

Example 10

The scale on a map is 1 : 500 000. If the distance from Campbelltown to Sydney on the scale drawing is 9 cm, calculate the actual distance from Campbelltown to Sydney.

Solution

Scaled distance = 9 cm

Actual distance = 9 × 500 000 cm 500 000 times larger

= 4 500 000 cm

= 45 000 m 1 m = 100 cm

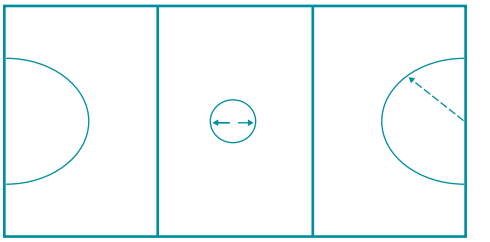
= 45 km 1 km = 1000 m

EXERCISE 13.03 ANSWERS ON P. 607

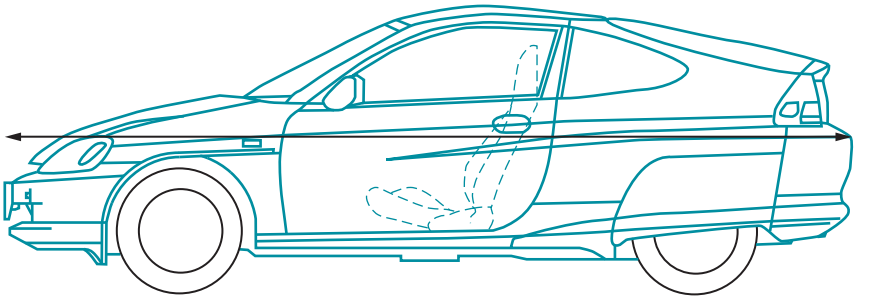
Scale drawings U F R

EXAMPLE 8

1 If the length of a netball court is 30.5 m, what scale has been used on this diagram?



2 What scale has been used on this diagram of a car if its actual length is 3.92 m?



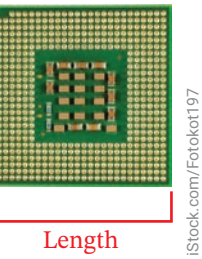
EXAMPLE 9

3 The picture of the Giant Brown Bull Ant has been enlarged. If its actual length is 25 mm, what scale has been used?



Nature Picture Library/Kim Taylor

4 A scale image of a microchip is shown. Determine the scale that has been used if its real length is 8 mm.



iStock.com/Fotokot197

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5 The Q1 Tower on the Gold Coast, Queensland, is 322 m high and is Australia's tallest building. What scale has been used in this photo if the scaled height is 40 mm?



Shutterstock.com/Steven Bostock

6 This photo of the Sydney Harbour Bridge has a scale of 1 : 9000. By measurement and calculation, find: R

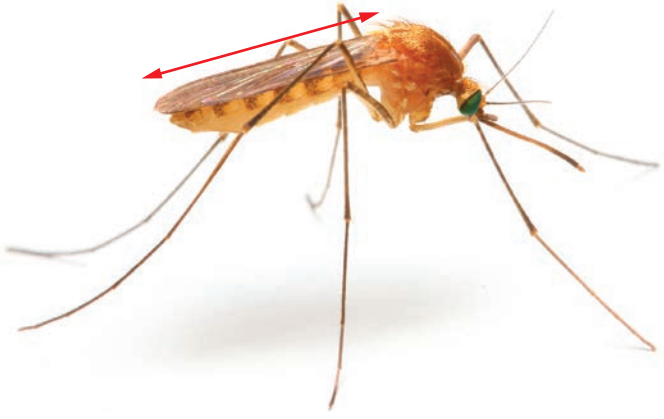
a the height of the top of the bridge above the water



Shutterstock.com/ErikSandstrom

b the length of the arch span of the bridge.

7 This is a photograph of a mosquito, at a scale of 15 : 1. Calculate, to the nearest 0.1 mm, the length of the wing. R



Shutterstock.com/Kletr

8 This box of cereal is 16 cm long and 22 cm tall. Construct a scale drawing of the front of the box, using a scale of 1 : 5.

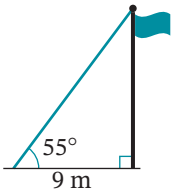


Shutterstock.com/Nor Gal

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9 **a** Construct a scale drawing of this flagpole using a scale of 1 : 300.

b Use the scale to calculate the actual height of the flagpole. R



EXAMPLE 10

10 The scale on a map of Australia is 1 : 10 000 000. If the distance from Perth to Adelaide on the map is 21.2 cm, calculate the actual distance in kilometres. PS R

11 Use the scale on this map to find, to the nearest kilometre, the distance from: R

a Nowra to Batemans Bay

b Bega to Moruya

c Mollymook/Ulladulla to Mogo

d Nowra to Eden.



Investigation

Floor plans

Construct a scale drawing of your bedroom or classroom.

Measure the dimensions of the room and each piece of furniture in it. Remember to measure the location of immovable objects such as windows, doors, wardrobes and whiteboards. Include a list of your measurements and the scale used. You may also do an Internet search of architectural symbols (bed, lounge, sink etc.) and include these in your room, where applicable.

Foundation Standard Complex

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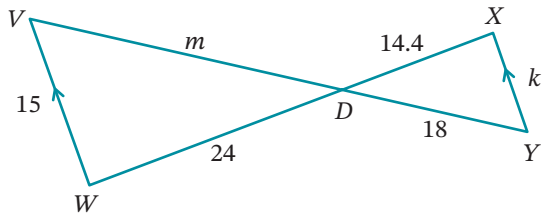
Power plus ANSWERS ON P. 607

1 **a** What information is shown in the diagram?

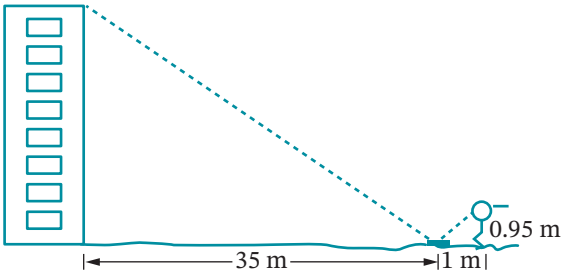
b Why is $\angle V = \angle Y$ and $\angle W = \angle X$?

c Show that $\triangle VWD \sim \triangle YXD$.

d Find the values of k and m.



2 Alexa places a mirror on the ground. She stands 1 m from the mirror and then bobs down until she can see the top of a building in the mirror. Her eyes are 95 cm above ground level when she can see the building. If the mirror is 35 m from the base of the building, what is the height of the building?



3 Is each statement true or false? Give reasons.

a All equilateral triangles are similar.

b All isosceles triangles are similar.

c All circles are similar.

d All rectangles are similar.

e All squares are similar.

f All regular pentagons are similar.

Worksheet
Investigating paper sizes

13.03

CHAPTER 13 REVIEW

Quiz
Language of maths 13

Language of maths

enlargement	image	matching	original
proportional	ratio	reduction	scale
scale drawing	scale factor	similar (~)	vertex/vertices

1 When an original figure is enlarged or reduced, what is the new figure called?

2 What is the symbol and meaning of 'is similar to'?

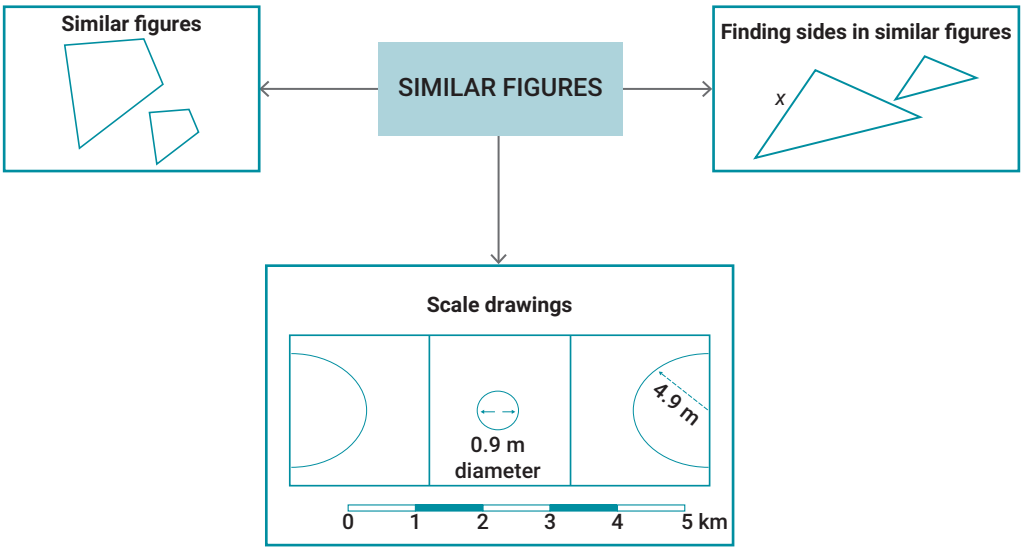
3 Which word from the above list is another name for 'corresponding'?

4 What happens to a shape that is changed by a scale factor of 3?

Topic summary

Worksheet
Mind map: Similar figures

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

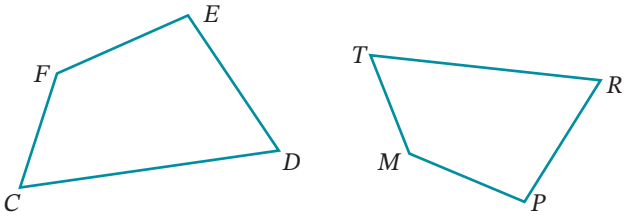


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TEST YOURSELF 13

ANSWERS ON P. 608

1 The quadrilaterals *FECD* and *TRMP* are similar. 13.01



a List all pairs of matching sides.

b List all pairs of matching angles.

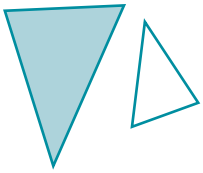
c Use the correct notation to write a congruency statement relating the 2 quadrilaterals.

2 By measurement, find the scale factor for each pair of similar figures. 13.01

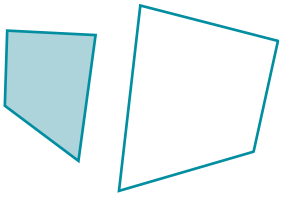
a



b



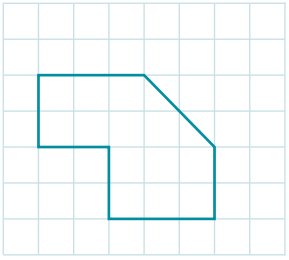
c



3 Copy each figure onto graph paper and draw its image if: 13.01

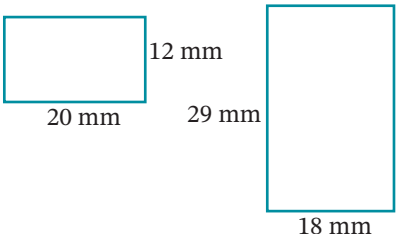
a scale factor = 3

b scale factor = $\frac{1}{2}$

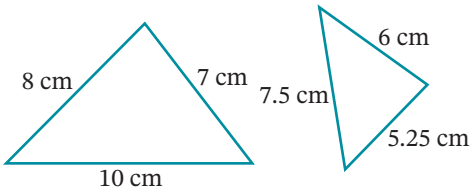


4 Test whether each pair of figures are similar. 13.02

a



b



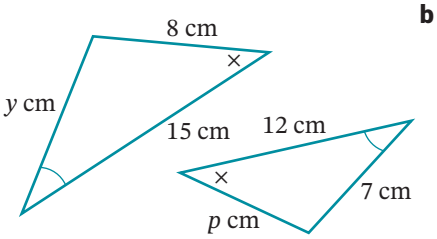
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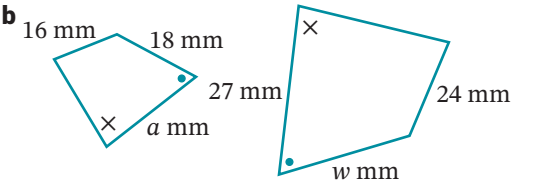
13.02

5 Find the value of each variable in each pair of similar figures.

a



b



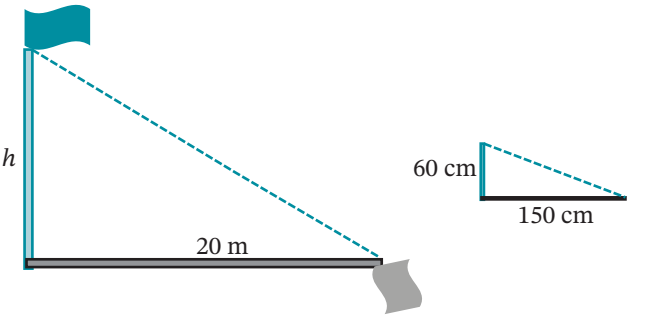
13.03

6 A flagpole's shadow is 20 m long. At the same time, a stick 60 cm tall casts a shadow 150 cm long.

What is the height, h, of the flagpole? Select **A**, **B**, **C** or **D**.

A 12 m **B** 50 m

C 8 m **D** 9 m

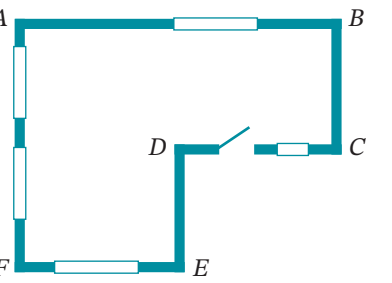


13.03

7 This floor plan of a holiday house is drawn to a scale of 1 : 200. What is the actual (internal) length of:

a AB?

b CD?



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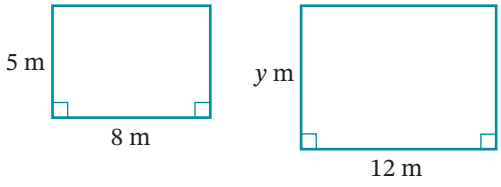
PRACTICE SET 4

ANSWERS ON P. 608

1 Test whether the point (2, 3) lies on each line. 11.04

a $2x + y = 4$ **b** $y = 2x - 1$ **c** $x + y = 5$

2 The rectangle on the left has been enlarged to make the rectangle on the right. 13.02



a What is the scale factor?

b Find the value of y.

3 The probability of Eva hitting a bullseye on a target is 0.2. Find the probability of Eva missing the bullseye. 12.01

4 A class was surveyed on whether its students could swim. The results are shown. PATH 12.04

	Can swim	Cannot swim
Boys	13	2
Girls	9	3

a Find the probability of randomly selecting a student who is:

i a boy

ii a boy or cannot swim

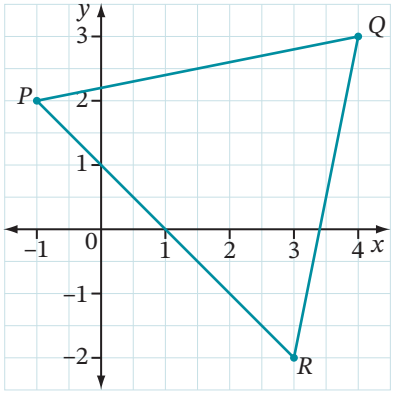
iii a girl who cannot swim

iv a girl or can swim, but not both.

b If a swimmer is selected at random, what is the probability that it is a boy?

5 The scale on a tourist map of Sydney is 2 cm = 500 m. If the scaled distance from Circular Quay station to the Opera House is 2.5 cm, what is the actual distance? 13.03

Questions **6** to **9** refer to this triangle whose vertices are $P(-1, 2)$, $Q(4, 3)$ and $R(3, -2)$.



6 Find the length of: 11.01

a side PQ as a surd

b side PR, correct to one decimal place.

7 Show that $\triangle PQR$ is isosceles. 11.01

8 Find the coordinates of the midpoint of side PR. 11.02

9 Find the gradient of: 11.03

a PQ **b** PR

Foundation Standard Complex

11.04

10 Graph $y = 4 - 2x$ on a number plane and write:

a its x-intercept **b** its y-intercept

12.02

11 Daniel rolled a biased die 90 times and the number 6 came up 24 times.

a What is the observed probability of rolling a 6?

b If the same die was rolled 400 times, what is the expected frequency of rolling a 6?

11.10

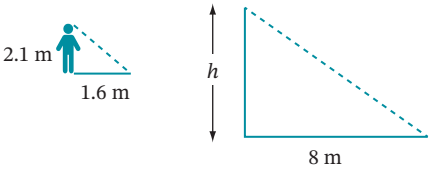
12 **a** Sketch the graph of $y = x^2 - 2$.

b Write the coordinates of the vertex.

c Is the graph concave up or concave down?

13.02

13 Joemon is 2.1 m tall and casts a shadow 1.6 m long while standing in the sun. At the same time, a flagpole casts a shadow 8 m long. How tall is the flagpole?



11.04

14 What is the equation of the horizontal line passing through (–2, 4)?

12.01

15 The numbers 1 to 30 are written on cards and the cards are placed in a bag. If one card is drawn at random from the bag, what is the probability that it is:

a a multiple of 5?

b greater than 10?

11.05

16 **a** Find the gradient and y-intercept of the line with equation $y = \frac{1}{2}x + 3$.

b Graph a line with gradient 2 and y-intercept –5 and write its equation.

PATH

11.09

17 The increase in pressure experienced by a scuba diver is directly proportional to their depth under the water. The increase in pressure at depth 40 m is 235 kPa (kilopascals).

a What increase in pressure is experienced at a depth of 32 m?

b At what depth, to the nearest metre, is the increase in pressure 735 kPa?

12.05

18 2 dice are rolled together.
Copy and complete the table to show the sample space.

		1st die					
		1	2	3	4	5	6
2nd die	1						
	2					5, 2	
	3						
	4		2, 4				
	5						
	6						

Find the probability of rolling:

a a 3 and a 6 (in any order)

b at least one 2

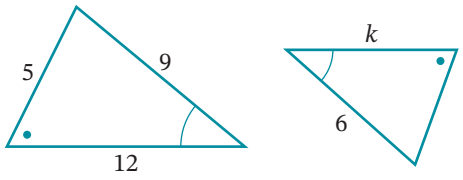
c 2 numbers the same

d a 4 or a 5, but not both

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19 What is the value of k if these triangles are similar? Select the correct answer **A**, **B**, **C** or **D**.



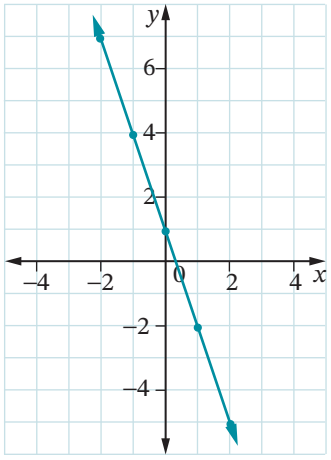
13.02

A 8 **B** 9

C 10 **D** 12

20 **a** Find the gradient of this line.

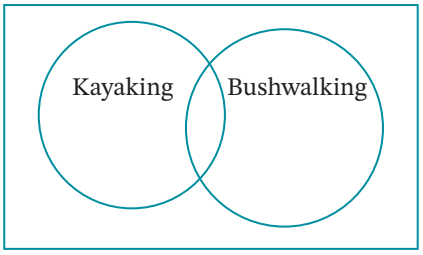
b Find the equation of this line.



11.06

PATH

21 40 students at a school camp could select kayaking or bushwalking as an activity, or both. A total of 35 students chose kayaking, 20 chose bushwalking, and some chose both. There weren't any students who chose neither.



12.03

a Copy and complete the Venn diagram representing these students.

b What is the probability that a student selected at random:

i chose kayaking and bushwalking?

ii chose kayaking or bushwalking, but not both?

22 Match each equation to the correct description of its graph.

11.10

a $y = x^2 + 9$

b $y = x + 9$

c $y = 9$

d $x = 9$

A straight line

B horizontal line

C parabola

D vertical line

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